Present Day, Present Time: AI Exposure & Labor Market Risk

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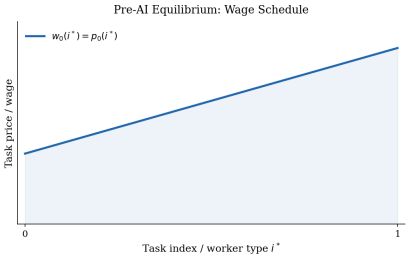
The Question

November 2022: ChatGPT launches.

Millions of people start using AI to do things that *people get paid to do*.

Some workers are much more exposed than others.

Did it cost them?



Workers differ in how substitutable their tasks are

A Lesson from Trade Economics

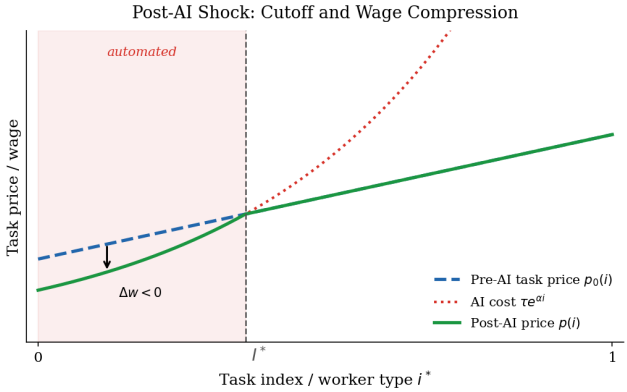
Offshoring (2000s): companies shipped tasks abroad when foreign labor was cheaper

- **Grossman & Rossi-Hansberg (2008):** it's not whole jobs that get offshored — it's specific *tasks*
- **Ottaviano, Peri & Wright (2013):** different workers face different cutoffs — not everyone equally at risk
- **AI is the same story, new technology:**

  Instead of cheap foreign workers — a cheap *machine*.
  "Offshoring" your tasks to Claudeland, GPTopia, Geminia...

The Model

Key idea: rank every cognitive task by how easily AI can do it



- **Left of I^*:** routine tasks — AI undercuts

  $\rightarrow$ immediate wage pressure

- **Right of I^*:** complex tasks — humans keep the edge

- As AI improves, I^* shifts right — more workers get exposed over time

What the Model Predicts

Workers to the left of l^* face two immediate pressures:

Wage compression

Firms can switch to AI $\rightarrow$ workers must accept lower pay to stay employed

Displacement

If the compressed wage drops below what you're willing to accept $\rightarrow$ you exit

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This captures the **immediate substitution effect** *only — longer-run channels (augmentation, output expansion, new tasks) could go the other way.*

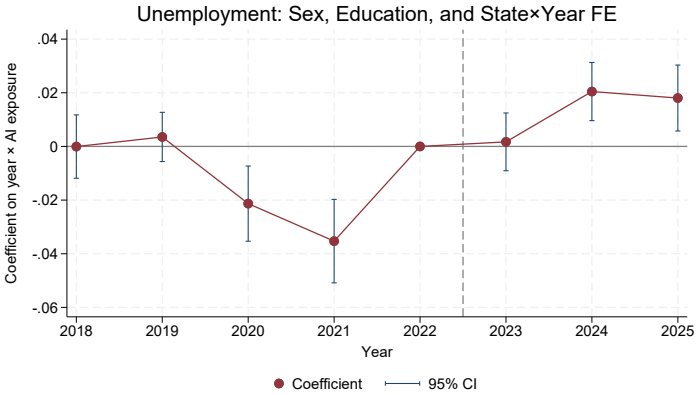
The Strategy

$$Y_{it} = \alpha + \sum_{t \neq 2022} \beta_t \left(\mathbf{1}[t] \times \mathrm{AIExp}_i \right) + X_{it}' \Gamma + \varepsilon_{it}$$

X_{it}: sex, education, state×year FEs
SEs clustered by occupation Baseline: 2022
($\beta_{2022} \equiv 0$)

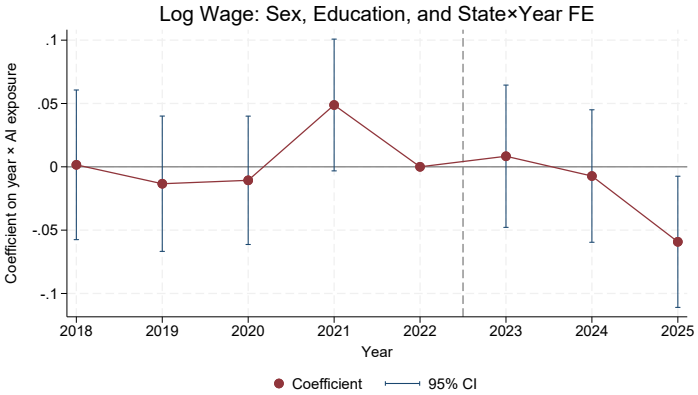
- **Data:** CPS, 500K+ workers, 2018–2025
- **AI exposure:** Eloundou et al. (2024), occupation-level LLM score
- **Strategy:** high vs. low exposure, before vs. after ChatGPT
- **Model says:** β_t diverges post-2022

Result 1: Unemployment



- Flat pre-trends before 2022 — groups are comparable
- **Post-2022:** high-exposure workers show a rising unemployment gap
- Effect holds across all specifications

Result 2: Wages



- Flat pre-trends — consistent baseline
- **Post-2022:** wages compress for high-exposure workers
- Non-college workers bear the brunt

Thank you!

Questions welcome.

Bottom line:

AI is like offshoring —
some workers face immediate
wage and employment risk

The data says:

Yes, high-exposure workers
are already seeing it —
unemployment *and* wages

Backup: The Dirty Work

Formal model, derivations, and further discussion of I^*

Model Setup (Formal)

- **Tasks** indexed $i \in [0, 1]$ by AI susceptibility
  - Low i: routine cognitive — easy for AI — High i: complex — hard for AI *currently*
- **CES Production**:

$$Y = \left[\int_0^1 y(i)^{\frac{\sigma-1}{\sigma}} \, di \right]^{\frac{\sigma}{\sigma-1}}$$

- **Workers:** type i^* specializes in task i^*; supply of task i is $f(i)$ (fixed, inelastic)
- **AI cost** of task i: $c^{\mathrm{AI}}(i) = \tau \cdot e^{\alpha i}$
  - Lower τ = more capable AI; $e^{\alpha i}$ ensures complex tasks remain costly for AI
- Competitive firms; output price P_Y fixed (small open economy)

Pre-AI Equilibrium: Derivation

Firm cost minimization gives task demand:

$$y^d(i) \;=\; Y \cdot \left(\frac{P_Y}{p_0(i)}\right)^{\sigma}$$

Market clearing $\left(y^d(i) = f(i)\right)$ pins down task price:

$$p_0(i) \;=\; P_Y \cdot \left(\frac{Y}{f(i)}\right)^{1/\sigma}$$

Solve for Y_0 — plug $y(i) = f(i)$ back into the CES aggregator:

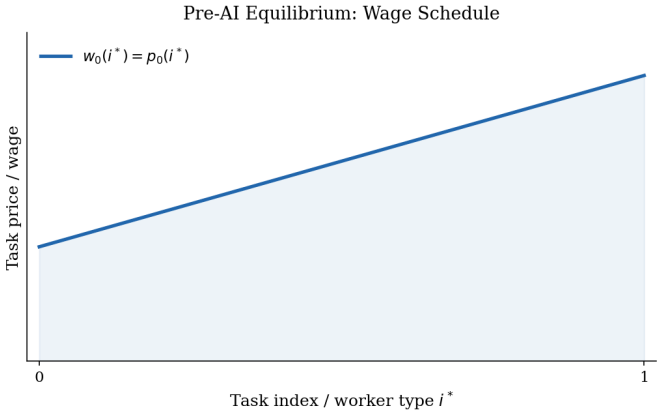
$$\boxed{Y_0 \;=\; \left[\int_0^1 f(i)^{\frac{\sigma-1}{\sigma}}\, di\right]^{\frac{\sigma}{\sigma-1}}}$$

- Y_0 is fully determined by worker distribution f — not by prices
- Y_0 is the output level held fixed in the short-run AI shock analysis

Pre-AI Equilibrium: Wage Schedule

Under full specialization, each worker earns their task price:

$$w_0(i^*) \;=\; p_0(i^*) \;=\; P_Y \cdot \left(\frac{Y_0}{f(i^*)}\right)^{1/\sigma}$$



- Each worker earns the competitive price of their task
- Scarce types earn more; abundant types earn less

Upward slope for illustration only — no shape restriction on p_0

Immediate Effects: What We Do and Don't Model

Hold $Y = Y_0$ fixed — immediate impact before the economy adjusts

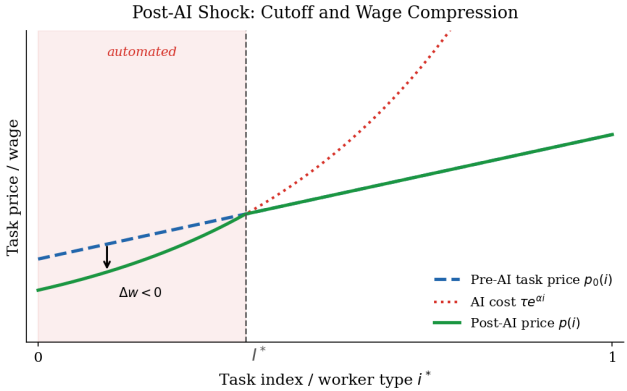
Isolates the **substitution channel** only. Longer-run channels set aside:

- **Augmentation** — AI complements some workers, raising productivity over time
- **Output expansion** — cheaper production eventually raises Y beyond Y_0
- **Task creation** — AI creates new tasks over time (Acemoglu & Restrepo 2019)

Immediate task price once AI enters:

$$p(i) = \min\big(p_0(i),\ \tau e^{\alpha i}\big) \qquad \Rightarrow \qquad \text{cutoff } I^* :\ \tau e^{\alpha I^*} = p_0(I^*)$$

Immediate Effects: Cutoff and Wages



- $i < I^*$: AI immediately undercuts → wage compression

$$\Delta w = \tau e^{\alpha i^*} - p_0(i^*) < 0$$

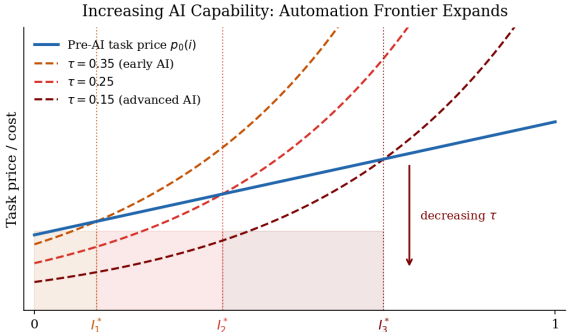
- $i \geq I^*$: no direct price pressure in the short run
  - ↪ output expansion raises their wages over time
- Compressed wage < reservation → **exit employment**

Further Discussion of I^*

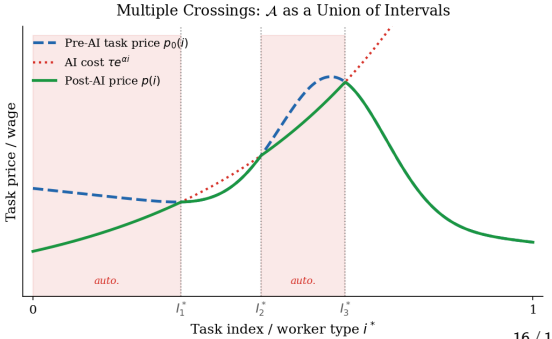
As AI improves ($\tau \downarrow$), I^* rises

$$\frac{\partial I^*}{\partial \tau} < 0$$

The automation frontier expands toward more complex tasks.



Non-monotone $p_0(i)$: multiple cutoffs

$$\mathcal{A} = [0, I_1^*] \cup [I_2^*, I_3^*] \cup \cdots$$

Wage compression applies to all $i^* \in \mathcal{A}$; exposed workers need not be at the low end.